# SOP 01 | Patch an Existing Oracle 19c Database

*Scenario 1: upgrade the currently installed and operational Oracle 19.3 Enterprise Edition to Windows Bundle Patch 19.33.*

Document control | Version 1.0 | Issued 04 November 2026 | Windows x64 | Single-instance Enterprise Edition

USE THIS SOP FOR YOUR CURRENT SERVER. Database ORCLCDB, PDB ORCLPDB1 and APP_DATA already exist. Do not rerun setup.exe, DBCA, or create a second database.

| Item | Approved baseline / target |
|---|---|
| Operating system | Windows Server 2025 Standard Evaluation, build 26100 (lab) |
| Oracle edition | Oracle Database 19c Enterprise Edition, single instance |
| Oracle Home | C:\app\oracle\product\19.0.0\dbhome_1 |
| Oracle Base | C:\app\oracle |
| Database / service | ORCLCDB / ORCLPDB1; listener LISTENER TCP 1521 |
| Before / after | Oracle 19.3.0.0.0 → Oracle 19.33 (Windows Bundle Patch) |
| Scope | Standalone filesystem deployment; no Grid/RAC/ASM |

## Decision and stop conditions

- This is an in-place Oracle Home patch. Plan an outage; database and listener will be stopped.
- Patch data dictionary only AFTER binary patching and restart. Backup first, then patch, then datapatch.
- Existing 19.3 on Windows Server 2025 remains unsupported until 19.27+ is installed. Do not use it for production before successful certification and other required controls.
- If support patch downloads are unavailable, STOP at download gate: do not attempt third-party/altered ZIP files.

## Required files: exact selection

| Artifact | What to download | Required? |
|---|---|---|
| Oracle DB base image | V982656-01.zip (Oracle 19.3 Windows x64 Database Home). For a new installation only. | Scenario 2 only |
| Windows Database Bundle Patch 19.33 | My Oracle Support patch ID 39898799; select Microsoft Windows x64 (64-bit), Oracle Database 19c, Oct 2026. | YES |
| OPatch utility | My Oracle Support patch ID 6880880; choose compatible 19c Windows x64 build meeting bundle README minimum. | YES |
| Oracle Client | V982658-01.zip; separate client installer, NOT required for local database server patching. | NO |
| OJVM / Data Pump patches | OJVM 39803650 and Data Pump 40057600 are conditional, not part of baseline procedure. Verify installed component and exact Windows applicability/readmes. | Conditional |
| Grid Infrastructure / Linux RU | Do not download the Linux DB RU 39915792 or Windows Grid/RAC bundle for this standalone database. | NO |

Do not assume the actual patch ZIP filename. Example naming patterns are p39898799_190000_MSWIN-x86-64.zip and p6880880_190000_MSWIN-x86-64.zip; use the filename presented by the entitled Oracle download page. The patch ID, product, platform, release and README must all match.

- Download through https://support.oracle.com/ using a valid support CSI; an ordinary free Oracle account generally cannot download patches.
- Keep original ZIPs unchanged in C:\OraclePatches\downloads; extract into a separate staging folder. Never extract a bundle directly into ORACLE_HOME.
- Obtain and read the bundle patch README before using any apply command. The README overrides generic commands in this SOP.

**PAUSE / CHECKPOINT | G0 — Download authenticity and applicability**

GO when: All required archives are present; Oracle patch page and README identify the Windows x64 DB 19.33 bundle for 19c, patch ID 39898799; OPatch platform and minimum version are confirmed.
STOP when: Do not download from mirror sites, use a Linux RU, or apply any unverified patch.

## Step 1 — Baseline the running database

Run on the database server and retain outputs with your change record.

## PowerShell conventions and staging

Use PowerShell opened with “Run as administrator” unless SQL> or RMAN> is explicitly shown. Commands beginning with SQL> belong in SQL*Plus; do not paste them directly into PowerShell. All paths are examples grounded in this Oracle Home.

```
$OH = 'C:\app\oracle\product\19.0.0\dbhome_1'
$env:ORACLE_HOME = $OH            # current session only
$env:ORACLE_SID  = 'ORCLCDB'      # current session only
$env:PATH = "$OH\bin;$OH\OPatch;$env:PATH"
$stage = 'C:\OraclePatches\19.33'
New-Item -ItemType Directory -Force -Path $stage | Out-Null
Get-CimInstance Win32_OperatingSystem |
  Select-Object Caption,Version,BuildNumber,OSArchitecture
Test-Path "$OH\OPatch\opatch.bat"
```

Use the downloaded ZIP location appropriate to the machine. Confirm file hashes against My Oracle Support details when provided, not against an unrelated ZIP with a different package layout.

```
Get-FileHash 'C:\OraclePatches\downloads\<actual-patch-file>.zip' -Algorithm SHA256
```

```
& "$OH\OPatch\opatch.bat" version
& "$OH\OPatch\opatch.bat" lsinventory
```

**PAUSE / CHECKPOINT | G1 — Baseline and inventory**

GO when: Oracle Home exists and maps to the intended registered Home; OPatch reports its version and a readable Oracle inventory.
STOP when: Inventory errors, wrong Oracle Home, or missing OPatch must be resolved before applying anything.

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT con_id,patch_id,action,status,description
FROM cdb_registry_sqlpatch
ORDER BY action_time;
SELECT comp_id,status FROM dba_registry WHERE comp_id='JAVAVM';
EXIT;
```

**PAUSE / CHECKPOINT | S1-01 — Pre-patch state**

GO when: ORCLCDB is READ WRITE, CDB=YES; ORCLPDB1 is READ WRITE; current patch and OJVM presence documented; no unrelated service outage.
STOP when: If the database is already impaired, resolve baseline faults before patching.

## Step 2 — Back up and prove recoverability

Use a separate disk/backup repository with sufficient free capacity; D: is a placeholder, not a confirmed volume. A powered-off VM snapshot may supplement but does not replace a database backup. Since ARCHIVELOG is enabled, RMAN can capture the entire CDB and its PDBs.

```
# In elevated PowerShell, adapt to an EXISTING separate backup volume.
Test-Path 'D:\OracleBackups'
# If the D: backup volume exists and is approved:
New-Item -ItemType Directory -Force -Path 'D:\OracleBackups\PrePatch' | Out-Null
& "$OH\bin\rman.exe" target /
```

```
-- RMAN> prompt
CONFIGURE CONTROLFILE AUTOBACKUP ON;
BACKUP DATABASE PLUS ARCHIVELOG
  FORMAT 'D:\OracleBackups\PrePatch\%U.bkp'
  TAG 'PRE_1933';
BACKUP CURRENT CONTROLFILE FORMAT 'D:\OracleBackups\PrePatch\control_%U.bkp';
BACKUP SPFILE FORMAT 'D:\OracleBackups\PrePatch\spfile_%U.bkp';
LIST BACKUP SUMMARY;
RESTORE DATABASE VALIDATE;
EXIT;
```

The RMAN format string above represents a Windows file path; enter with single path separators in the RMAN prompt as D:\OracleBackups\PrePatch\%U.bkp. Verify backup files exist, the RMAN log records success and RESTORE DATABASE VALIDATE completes. Also retain Oracle Home/configuration and patch inventory backups and a tested restore plan.

**PAUSE / CHECKPOINT | S1-02 — Recovery ready**

GO when: Successful RMAN backup of CDB/PDB and archived redo, controlfile autobackup, restore validation, protected backup destination and a documented rollback/recovery plan.
STOP when: Do not patch a populated database without a usable backup and recovery strategy.

## Step 3 — Download, stage and update OPatch

## Update the OPatch utility

1. Extract the downloaded OPatch ZIP into C:\OraclePatches\opatch_extract (NOT the database patch folder). Verify that it contains a top-level OPatch directory.

```
$opatchZip = 'C:\OraclePatches\downloads\<actual-6880880-zip>.zip'
$extract = 'C:\OraclePatches\opatch_extract'
Expand-Archive -LiteralPath $opatchZip -DestinationPath $extract -Force
Test-Path "$extract\OPatch\opatch.bat"
```

2. Back up the old OPatch folder; replace it with the extracted OPatch directory. Avoid merging folders, as stale files could remain. Ensure no Oracle patch utility is running.

```
Rename-Item -LiteralPath "$OH\OPatch" -NewName 'OPatch_before_1933'
Copy-Item -LiteralPath "$extract\OPatch" -Destination "$OH\OPatch" -Recurse
& "$OH\OPatch\opatch.bat" version
```

3. Compare OPatch version with the minimum version in the 19.33 Windows bundle README. A higher compatible version is acceptable if the README permits it.

**PAUSE / CHECKPOINT | G2 — OPatch readiness**

GO when: New OPatch folder contains opatch.bat; “opatch version” works; version meets the 19.33 README; previous OPatch folder remains available for recovery.
STOP when: If opatch version fails, stop; do not attempt an Oracle database bundle patch.

## Step 4 — Plan outage and stop database and services

4. Notify users and disconnect applications. Close SQL Developer, SQL*Plus, RMAN, schedulers and other processes that load DLLs from this Oracle Home.

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SHUTDOWN IMMEDIATE;
EXIT;
```

```
Stop-Service -Name OracleOraDB19Home1TNSListener
Stop-Service -Name OracleServiceORCLCDB
Get-CimInstance Win32_Service |
  Where-Object Name -match 'Oracle' |
  Select-Object Name,State,StartMode
Get-Process sqlplus,rman,tnslsnr,oracle -ErrorAction SilentlyContinue |
  Select-Object Name,Id,Path
```

Inspect OracleRemExecServiceV2, OracleVssWriterORCLCDB and other services. Stop any service locking the Oracle Home if required by the bundle README. Do not force-kill unknown processes or alter unrelated Windows services.

**PAUSE / CHECKPOINT | S1-03 — Maintenance outage achieved**

GO when: Database shutdown completed cleanly; OracleServiceORCLCDB and Oracle listener service are STOPPED; no active processes have Oracle Home files locked; maintenance window remains approved.
STOP when: If stop fails or Oracle Home files remain locked, investigate before applying the bundle.

## Step 5 — Patch Oracle binaries

## Unpack the Windows Database Bundle Patch

```
$bundleZip = 'C:\OraclePatches\downloads\<actual-39898799-zip>.zip'
$stage = 'C:\OraclePatches\19.33'
Expand-Archive -LiteralPath $bundleZip -DestinationPath $stage -Force
Get-ChildItem -LiteralPath $stage -Recurse -Filter README* |
  Select-Object FullName
Get-ChildItem -LiteralPath $stage -Recurse -Directory |
  Where-Object Name -eq '39898799' | Select-Object FullName
```

An Oracle patch ZIP commonly extracts a folder named 39898799. Your actual layout may differ. Locate the patch-root directory containing etc, files and inventory and set $PatchDir to that exact location. Do not guess the folder from the ZIP filename.

```
$PatchDir = 'C:\OraclePatches\19.33\39898799' # adjust after inspection
Test-Path "$PatchDir\etc"
Test-Path "$PatchDir\files"
& "$OH\OPatch\opatch.bat" prereq CheckConflictAgainstOHWithDetail -ph $PatchDir
```

**PAUSE / CHECKPOINT | G3 — Patch prerequisites**

GO when: README has been reviewed; the extracted root is correct; opatch prereq reports that conflict checks are satisfied

and no mandatory dependency is missing.
STOP when: Stop if there is a conflict, wrong platform, missing prereq, or failed inventory. Do not use -force to bypass conflicts.

## Apply the patch to the Oracle Home

The database services must be stopped for Scenario 1. Scenario 2 has no database instance yet, but any background Oracle process from the new Oracle Home must still be closed.

```
Set-Location -LiteralPath $PatchDir
& "$OH\OPatch\opatch.bat" apply
```

Follow the on-screen prompts and the exact 19.33 Windows README instructions. The generic command shown here is appropriate only when README specifies standard in-place OPatch application. Never terminate the patch process halfway through.

```
& "$OH\OPatch\opatch.bat" lsinventory
& "$OH\OPatch\opatch.bat" lspatches
```

**PAUSE / CHECKPOINT | G4 — Binary patch successful**

GO when: OPatch ends with success and inventory clearly lists the expected 19.33 Windows Database Bundle Patch 39898799 (or its README-documented constituent patch identifiers).
STOP when: If OPatch fails or inventory lacks the target patch, STOP before running DBCA or datapatch. Inspect the OPatch log and use Oracle's documented recovery procedure.

## Step 6 — Start the CDB, open the PDB and run datapatch

## Apply SQL-level updates using datapatch

OPatch changes binaries; datapatch updates the database data dictionary and patch registry. Because ORCLCDB and ORCLPDB1 already exist, this stage is mandatory when the Windows bundle README includes SQL patch actions.

```
$env:ORACLE_SID = 'ORCLCDB'
Start-Service -Name OracleServiceORCLCDB
Start-Service -Name OracleOraDB19Home1TNSListener
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
-- SQL> prompt; do NOT paste into PowerShell
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
-- Open ORCLPDB1 only if it is not READ WRITE:
-- ALTER PLUGGABLE DATABASE ORCLPDB1 OPEN READ WRITE;
EXIT;
```

After exiting SQL*Plus, discover the installed datapatch launcher. On Windows this is normally datapatch.bat; run the version actually installed in OPatch.

```
Get-ChildItem "$OH\OPatch\datapatch*" | Select-Object Name
& "$OH\OPatch\datapatch.bat" -verbose
```

Confirm that SQL patch actions for root, seed and ORCLPDB1 report SUCCESS or document a README-permitted state. If an individual PDB was closed, open it and rerun datapatch for it as directed.

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SET LINES 180
COLUMN description FORMAT A55
SELECT con_id,patch_id,action,status,description
FROM cdb_registry_sqlpatch
ORDER BY action_time,con_id;
SHOW PDBS;
SELECT comp_id,status FROM dba_registry WHERE status <> 'VALID';
```

**PAUSE / CHECKPOINT | G5 — SQL patch successful**

GO when: datapatch logs have no unresolved failure; the expected patch SQL actions are SUCCESS for relevant containers; ORCLPDB1 is READ WRITE and components are valid or reviewed.
STOP when: If datapatch reports failure, stop further configuration. Read logs under C:\app\oracle\cfgtoollogs\sqlpatch and follow Oracle's recovery instructions.

## Step 7 — Acceptance checks and sign-off

## Final acceptance and service tests

```
& "$OH\OPatch\opatch.bat" lsinventory
Get-CimInstance Win32_Service |
  Where-Object Name -match 'Oracle' |
  Select-Object Name,State,StartMode
& "$OH\bin\lsnrctl.exe" status LISTENER
Test-NetConnection 127.0.0.1 -Port 1521
```

```
& "$OH\bin\sqlplus.exe" / as sysdba
```

```
SELECT instance_name,version_full,status FROM v$instance;
SELECT name,open_mode,log_mode,cdb FROM v$database;
SHOW PDBS;
SELECT con_id,patch_id,action,status
FROM cdb_registry_sqlpatch
ORDER BY action_time DESC,con_id;
```

Test a real service connection, not only TNSPING. SQL*Plus password prompts may need quoted input when a password contains @; do not place passwords in command lines, scripts or runbook evidence.

```
& "$OH\bin\sqlplus.exe" /nolog
```

```
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- Enter administrative password at prompt; if it contains @,
-- enclosing password in double quotes worked in this lab.
SHOW CON_NAME;
SHOW USER;
```

**PAUSE / CHECKPOINT | G6 — Release acceptance**

GO when: OPatch inventory reports the intended bundle, SQL registry shows SUCCESS in required containers, ORCLPDB1 is READ WRITE, Oracle services are running, listener responds on 1521 and authenticated SQL*Plus connects.
STOP when: If any acceptance test fails, do not sign off the change. Record full error and relevant OPatch/sqlpatch/listener logs.

## Rollback and evidence

If G4 fails, keep the database down, retain OPatch logs and follow the exact README rollback or restoration instructions. If G5 fails, avoid mixing binary and SQL levels; consult datapatch logs and Oracle Support before any rollback. Do not blindly restore only one of binaries/datafiles from a different point in time.

Retain: approved change request, ZIP filenames and SHA256, MOS README, pre/post OPatch inventory, prepatch RMAN backup and validation, shutdown/startup evidence, datapatch log, CDB_REGISTRY_SQLPATCH output, listener and application connectivity tests.

## Source references and operator notes

**Oracle: Windows OS minimum version (19.27 for Server 2025):** https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/operating-system-checklist-oracle-database-installation-microsoft-windows.html

**Oracle Database 19c Windows downloads:** https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html

**Oracle Database Patch Maintenance 19c:** https://docs.oracle.com/en/database/oracle/oracle-database/19/dbptc/index.html

**Oracle: database creation and datapatch:** https://docs.oracle.com/en/database/oracle/oracle-database/19/admin/creating-and-configuring-an-oracle-database.html

**My Oracle Support - authenticated patch downloads and READMEs:** https://support.oracle.com/

**Oracle Windows DB 19.33 patch ID cross-check (third-party tracker, NOT a substitute for MOS):** https://www.markusdba.de/2026/11/01/oracle-database-19c-auf-windows-ru-patch-nummern/

Authority rule: My Oracle Support patch record and its README govern platform, prerequisites, specific operations and rollback. This runbook provides a controlled sequence, not a replacement for Oracle's patch-specific instructions.